High Postoperative Blood Pressure After Cardiac Surgery Is Associated With Acute Kidney Injury and Death

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Objectives: **Gaps and uncertainty exist regarding the understanding of optimal clinical goals for perioperative (ie, preoperative, intraoperative, and postoperative) blood pressure (BP) management in patients undergoing cardiac surgery and the consequences of achieving or failing to achieve those goals. In this setting, it is understood that preoperative hypertension is predictive of poor postoperative outcomes, with a growing appreciation that current, clinically acceptable changes in intraoperative BP also may be associated independently with adverse short- and long-term outcomes. In contrast, the impact of postoperative BP on outcomes after cardiac surgery remains less clear.**

Design: **This study was a retrospective outcome analysis.**

Setting: **The study included all cardiac surgery patients cared for at a single institution over a 7-year period. Consequences of the success or failure of meeting postoperative BP targets on medical outcomes and health resource utilization were evaluated.**

Results: **The study comprised 5,225 patients. Hypertensive postoperative patients experienced a higher in-hospital mortality rate compared with matched-case normotensive patients (4.97% *v* 1.32%, $p < 0.001$) and a longer hospital stay ($p = 0.024$). In hypertensive patients, serum creatinine levels from postoperative day 1 through postoperative day 7 were increased compared with baseline and postoperative renal dysfunction according to the Kidney Disease: Improving Global Outcomes criteria occurred significantly more often (25.3% *v* 19.7%, $p = 0.027$).**

Conclusions: **Postoperative hypertension is associated with compromised outcome as reflected by higher mortality, longer length of stay, and higher incidence of renal dysfunction.**

KEY WORDS: high postoperative blood pressure, cardiac surgery, acute kidney injury

WORLDWIDE AN ESTIMATED 1 in 3 adults experiences hypertension (HTN), with the total number of affected people estimated to be more than 1 billion.[1–3] When subjected to the stresses of cardiovascular surgery, patients with pre-existing HTN are subject to wide swings in intraoperative blood pressure (BP) and are at increased risk of adverse short- and long-term outcomes.[4–9]

Intraoperative HTN, independent of pre-existing HTN, also is common during cardiac surgery, and its management can affect outcomes. Importantly, intraoperative HTN occurs in patients without any prior history of HTN.[10–14] The etiology of intraoperative HTN is multifactorial and mechanistically discrete from that of nonsurgical hypertension. Postoperative HTN for up to 48 hours post-procedure also is common after cardiac (and noncardiac) surgery and is related to a durable increase in sympathetic tone and ongoing fluid mobilization and shifts.[10,15,16]

Defining a "target" BP during the intraoperative or postoperative period is a routine part of anesthesia and cardiac surgical patient care, yet there is surprisingly little objective evidence on the appropriate clinical goals. Because of comorbidities and the magnitude of acute physiologic trespass, intraoperative BP management in the cardiac surgical patient remains an area of active best-practice management investigation. In addition, due to diverse patient demography, the plethora of coexisting conditions, and the wide variety of underlying contributors to postoperative BP alterations, there is no single established definition or standard of care for the treatment of postoperative HTN.[17–19] As a result, postoperative BP management decisions usually are based on provider knowledge, experience, and beliefs and are subject to substantial regional and geographic practice variation. Reviews and consensus panels have attempted to address BP management recommendations across different settings and types of surgery, including cardiac surgery. In the absence of formal guidelines, this literature has been considered the best source of information available.[18,20]

The impact of postoperative BP problems on outcomes in cardiac surgery patients remains poorly understood in part due to the fact that the diagnosis of postoperative HTN, albeit common, varies according to criteria used for its definition.[14,17,21]

Acute kidney injury (AKI) after cardiac surgery has an incidence rate of 2% to 8% and confers significant morbidity, increased resource utilization, and greater short- and long-term mortality.[22–25] Known risk factors for AKI after cardiac surgery are expansive and include both poorly controlled preoperative

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and intraoperative BP.[9,25,26] The specific contribution of postoperative BP and BP management on the development or progression of AKI and outcomes is less well-understood.

To better understand how postoperative BP affects short- and long-term surgical morbidity and mortality, the authors tested the hypothesis that postoperative HTN would be associated independently with increased mortality and adverse renal and health economic outcomes after cardiac surgery. Using a comprehensive perioperative electronic medical record database, the authors conducted a retrospective analysis of cardiac surgery patients cared for at a tertiary care hospital over a 7-year period.

METHODS

This study was approved by the federal data protection officer and the ethics commission of the Charité - Universitätsmedizin Berlin (EA1/034/13). The need for patient consent was waived due to the retrospective nature of the study, and therefore no intervention was performed. Clinical data from patients who underwent coronary artery bypass grafting surgery, valve surgery, or a combination of the 2 between 2006 and 2012 were extracted from the electronic patient data management systems (COPRA, Sasbachwalden, Germany, and SAP, Walldorf, Germany). Postoperatively, all patients were admitted to an intensive care unit (ICU). Excluded from the study were patients younger than 18 at the time of surgery, patients with a preoperative serum creatinine level $\geq$1.5 mg/dL, and patients who underwent preoperative hemodialysis.

Invasive blood pressure measurements via an indwelling arterial catheter were captured electronically. Throughout the study, data were captured progressively and recorded continuously in all patients for up to 24 hours after ICU admission or until discharge from the ICU, whichever occurred first. Captured blood pressures represented the mean value of invasive blood pressure measurements over a 30-minute period. Systolic blood pressure values above 250 mmHg or below 50 mmHg and diastolic blood pressure values above 200 mmHg and below 20 mmHg were regarded as erroneous and were excluded from the analysis. Data from monitoring devices automatically imported into the data management system were saved to the medical record after verification by a clinical care provider.

On arrival to the ICU, BP for all patients was managed in accordance with established standard operating procedures.[27] At Charité hospital the protocol requires that 130 mmHg—the European Society of Cardiology definition of arterial HTN—be used as the upper limit of the target range for normal BP, defined to be between 100 and 130 mmHg systolic.[28] The area under the curve (AUC) for excursion above and below this systolic blood pressure (SBP) target range (100 to 130 mmHg) therefore was used to define postoperative patient BP groupings for comparative analysis.

The authors defined patient groups as normotensive, hypotensive, or hypertensive, using the cumulative systolic SBP AUC excursions outside the referenced target range. The area outside the predefined target range was calculated separately for both hypertensive and hypotensive excursions. The area above the target range was denoted as AUC_{hyper}, and the area below the target range was denoted as AUC_{hypo} (Fig 1). Patients with cumulative AUC_{hyper} greater than the third AUC_{hyper} quartile were classified as "hypertensive"; patients with cumulative AUC_{hypo} greater than the third AUC_{hypo} quartile were classified as "hypotensive"; and patients whose AUC_{hyper} and AUC_{hypo} remained below the respective third quartile were grouped as "normotensive." To be clear, patients themselves were not grouped by quartiles. Rather, the patients' hypotensive and hypertensive excursions—quantified separately using AUC_{hypo} and AUC_{hyper} criteria—were determined and grouped by quartiles. The third quartile cut of each AUC (ie, AUC_{hyper} and AUC_{hypo}) was used as a basis to classify patients as either hypertensive or hypotensive. As a result, patients were classified as hypertensive when their AUC_{hyper} exceeded the third quartile. The third quartile cut point then was examined further using sensitivity analysis with different thresholds for defining hypertension (see the Statistical Analysis section).

The focus of this study was the comparison of outcome measurements between hypertensive and normotensive groups. In-hospital mortality, length of stay, time of mechanical ventilation, chest tube drainage and urine output, postoperative serum creatinine level, and renal dysfunction according to the

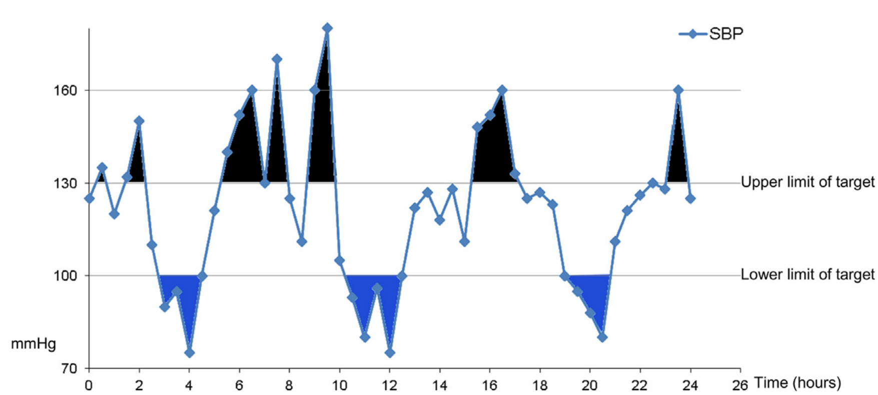

Fig 1. Definitions of AUC_{hyper} and AUC_{hypo}. AUC_{hyper} were assessed using the following 2 steps: for all patients who had never undergone a single measurement of SBP $>$130 mmHg during the 24 hour study period, AUC_{hyper} was set to 0, and for all patients who had at least 1 SBP measurement $>$130 mmHg during the 24-hour study period, AUC_{hyper} was calculated as illustrated above. AUC_{hyper} = sum of the *black* area; AUC_{hypo} = sum of *blue* area.

Kidney Disease: Improving Global Outcomes classification were compared.[29] The baseline serum creatinine level was defined as the last serum creatinine measurement before surgery.

Case-Control Matching

To eliminate the effect of confounding, 1:1 matched case (hypertensive) and control (normotensive) groups were established by identifying patients with similar baseline demographic and disease characteristics. The demographics and baseline covariates used for the matching were patient age, type and duration of surgery, preoperative glomerular filtration rate, presence of pre-existing diabetes mellitus, pre-existing arterial hypertension, and Acute Physiology and Chronic Health Evaluation II score on ICU admission.

Statistical Analysis

Descriptive analyses and statistical testing were performed using the R Project of Statistical Computing 3.0.1. Continuous variables were summarized using mean and standard deviation when normal distribution was confirmed using the Kolmogorov-Smirnov test; otherwise, median and interquartile ranges were provided. Categorical variables were summarized using frequencies and percentages. Statistical comparisons between groups were performed using the exact Mann-Whitney *U*-test for continuous variables. Chi-square tests were used for categorical data. Survival data were analyzed using Kaplan-Meier estimations and tested using the log-rank test between groups.

Odds ratios (OR) were determined for comparative mortality between hypertensive and normotensive groups at 4 different AUC_{hyper} cuts ($>$1,500; $>$3rd quartile; $>$2,000; $>$2,500 mmHg $\times$ min) to validate HTN (defined as excursion above 130 mmHg using different sensitivity criteria), which was associated with worse outcome. AUC_{hyper} was not used as a continuous variable due to the need to ensure adequate numbers for matching between groups. Calculated p values were performed from the log-rank test to demonstrate the strength of the finding. Sensitivity analyses to confirm the group effect of HTN on mortality were performed using a multivariate Cox proportional-hazards regression model and a multivariate logistic regression model adjusted for risk factors.

In addition, sensitivity analysis was performed for mortality outcome when the threshold for serum creatinine exclusion criteria varied between 1.3 mg/dL and 1.7 mg/dL. To test the potential impact of selection bias, additional 1:1 matched pairing (hypertensive to normotensive) was conducted by taking BP measurements into account that were recorded for at least 18 hours after ICU admission because data for $>$92% of patients were available for that time frame. Results of this subanalysis were compared with the complete data set that contained BP measurements for up to 24 hours postoperatively.

RESULTS

Patient Characteristics

A total of 6,263 patients 18 or older underwent cardiac surgery during the specified period and were screened for study inclusion. After excluding patients with preoperative hemodialysis (n = 203), preoperative serum creatinine levels above or equal to 1.5 mg/dL (n = 662), or missing values for the preoperative serum creatinine level (n = 312), a total of 5,225 patients were included for final analysis. The major demographic and baseline characteristics for the 5,225 patients are shown in Table 1 and include median age of 68 years, 71.2% male, 77.6% elective surgery, 57.5% coronary artery bypass grafting surgery, 77.6% coronary heart disease, and 72.3% preoperative HTN.

Immediately upon arrival to the ICU, approximately 90% of patients had SBP values less than 130 mmHg despite the fact that 3,764 (72%) patients were treated with intravenous antihypertensive medications within the first 24 hours of their postoperative ICU stay. Among the 5,225 patients, 891 (17%) were grouped as postoperative hypertensive; 3,158 (61%) were grouped as postoperative normotensive; and the remaining 1,176 (22%) were grouped as postoperative hypotensive. Of the patients grouped as postoperative hypertensive, 698 (78.3%) also had a history of preoperative HTN.

The data capture interval increased from 18 to 24 hours throughout the duration of the study as new data capture tools were used. Therefore, among all patients, BP measurements were captured continuously for at least 18 hours postoperatively for 4,588 (92%) patients; BP measurements were captured continuously for at least 20 hours postoperatively for 4,272 (86%) patients; and BP measurements were captured continuously for at least 22 hours postoperatively for 3,886 (78%) patients.

Baseline Profile of Matched Pairs

Of the 891 postoperative hypertensive patients, there were 604 (68%) 1:1 matched pairs in the normotensive group. Thus, demographics and baseline disease characteristics including age, type and duration of surgery, preoperative glomerular filtration rate, presence of pre-existing diabetes mellitus, pre-existing arterial HTN, and Acute Physiology and Chronic Health Evaluation II score on ICU admission for a total of 1,208 patients (604 in each group) were selected to establish 1:1 matched pairs between hypertensive and normotensive groups (Table 1). If not similar, characteristics for matched pairs were tolerated only when the difference was clinically in favor of the normotensive group (eg, hypertensive patients presented with a lower percentage of urgent/emergency surgery and lower Simplified Acute Physiology Score II and Therapeutic and Interventional Scoring System 28 scores than did respective patients in the normotensive group).

Outcome for Matched Pairs

Among the 1,208 patients in the 1:1 matched hypertensive and normotensive groups, 84% of the preoperative hypertensive patients and 78% of the preoperative normotensive patients received antihypertensive medication during the 24 hours after surgery while in the ICU. Postoperative hypertensive patients (regardless of preoperative HTN history) had a significantly increased risk for mortality (4.97% *v* 1.32%, unadjusted OR 3.83, 95% confidence interval [CI] 1.82-9.12, $p < 0.001$) and stayed 1 day longer in the hospital ($p = 0.024$; Table 2). In the Kaplan-Meier survival analyses, 30-day curves

Table 1. Patient Characteristics

	All Patients	Matched Patients		
	(n = 5,225)	Normotensive (n = 604)	Hypertensive (n = 604)	p Value
Basic characteristics				
Age (y)	68.0 (61.0-74.0)	71.0 (65.0-76.0)	71.0 (65.0-77.0)	0.472
Male	3,718 (71.2%)	403 (66.7%)	405 (67.1%)	0.951
Body mass index	26.8 (24.2-30.1)	26.9 (24.2-30.4)	27.0 (24.3-30.1)	0.942
Surgery-related data				
Preoperative ACEF score	1.26 (1.10-1.50)	1.30 (1.17-1.54)	1.28 (1.15-1.50)	0.274
Type of surgery				1.000
CABG	3,005 (57.5%)	397 (65.7%)	397 (65.7%)	
Valves	1,557 (29.8%)	144 (23.8%)	144 (23.8%)	
Both	663 (12.7%)	63 (10.4%)	63 (10.4%)	
Priority of surgery				0.008
Elective	3,401 (78.0%)	468 (77.5%)	494 (84.4%)	
Urgent	400 (9.18%)	53 (8.77%)	39 (6.67%)	
Emergency	558 (12.8%)	83 (13.7%)	52 (8.89%)	
Duration of anesthesia (min)	285 (235-330)	270 (230-310)	270 (230-310)	0.837
Duration of surgery (min)	195 (155-240)	185 (150-220)	180 (150-220)	0.679
Time on CPB (min)	87 (65-114)	78.5 (58.0-99.0)	78.0 (57.0-102)	0.951
Perioperative pRBC transfusion				
No. of patients	907 (24.8%)	160 (28.1%)	163 (29.2%)	0.720
No. of units in transfused patients	2.00 (1.00-2.00)	2.00 (1.00-2.00)	2.00 (1.00-2.00)	0.367
Scores on ICU admission				
APACHE II	18.0 (14.0-23.0)	18.0 (14.0-23.0)	18.0 (14.0-23.0)	0.762
SAPS II	34.0 (28.0-46.0)	35.0 (29.0-45.2)	34.0 (28.0-43.0)	0.030
TISS28	36.0 (32.0-40.0)	36.0 (33.0-40.0)	35.0 (32.0-39.0)	0.017
SOFA	6.00 (4.00-8.00)	6.00 (4.00-7.00)	6.00 (4.00-7.00)	0.273
Preexisting medical conditions				
Arterial hypertension	3,776 (72.3%)	484 (80.1%)	484 (80.1%)	1.000
Coronary heart disease	4,056 (77.6%)	492 (81.5%)	503 (83.3%)	0.450
Atrial fibrillation	1,579 (30.2%)	185 (30.6%)	191 (31.6%)	0.756
Hyperlipidemia	1,685 (32.2%)	180 (29.8%)	187 (31.0%)	0.707
Left heart failure (NYHA)				0.199
NYHA I	76 (1.45%)	8 (1.32%)	8 (1.32%)	
NYHA II	452 (8.65%)	53 (8.77%)	58 (9.60%)	
NYHA III	849 (16.2%)	88 (14.6%)	91 (15.1%)	
NYHA IV	680 (13.0%)	64 (10.6%)	40 (6.62%)	
NYHA unclassified	27 (0.52%)	3 (0.50%)	1 (0.17%)	
Left ventricular ejection fraction				0.081
>55	2,048 (53.5%)	258 (53.1%)	254 (56.1%)	
45-55	921 (24.1%)	119 (24.5%)	127 (28.0%)	
35-44	436 (11.4%)	62 (12.8%)	42 (9.27%)	
<35	423 (11.1%)	47 (9.67%)	30 (6.62%)	
IABP (preoperatively)	50 (0.96%)	2 (0.33%)	6 (0.99%)	0.287
Pulmonary hypertension	574 (11.0%)	49 (8.11%)	56 (9.27%)	0.540
Creatinine (preoperatively) (mg/dL)	0.98 (0.83-1.14)	1.01 (0.86-1.17)	1.00 (0.87-1.18)	0.791
Glomerular filtration rate preoperatively (mL/min/1.73 m^2)	76.1 (62.7-91.1)	71.9 (58.8-86.4)	71.6 (58.7-86.8)	0.921
Diabetes mellitus	2,031 (38.9%)	270 (44.7%)	271 (44.9%)	1.000
Peripheral vascular disease	920 (17.6%)	107 (17.7%)	103 (17.1%)	0.820
Chronic obstructive pulmonary disease	874 (16.7%)	98 (16.2%)	105 (17.4%)	0.644

Abbreviations: ACEF, age, creatinine, ejection fraction; APACHE II, Acute Physiology and Chronic Health Evaluation; CABG, coronary artery bypass grafting; CPB, cardiopulmonary bypass; IABP, intraaortic balloon pump; pRBC, packed red blood cells; SAPS II, Simplified Acute Physiology Score; SOFA, Sequential Organ Failure Assessment; TISS28, Therapeutic and Interventional Scoring System.

for matched postoperative normotensive and hypertensive groups differed significantly (p = 0.002; Fig 2). Levels of serum creatinine on postoperative day 1 were elevated in both groups, but the increase was higher in the hypertensive patient group (11.7% *v* 8.0% increase from baseline, p = 0.005). Within 1 week after surgery, the serum creatinine level in the normotensive patient group recovered to within 1.98% above baseline, whereas the serum creatinine level for patients in the hypertensive group remained at 6.62% above baseline (p = 0.005). Postoperative renal dysfunction occurred significantly

Table 2. Outcome Parameters of Matched Population

	Normotensive (n = 604)	Hypertensive (n = 604)	p Value
Selected outcome parameters			
Mortality (in hospital)	8 (1.32%)	30 (4.97%)	0.001
Length of stay on ICU (d)	5.00 (3.00-7.00)	5.00 (3.00-8.00)	0.119
Length of stay in hospital (d)	12.0 (8.00-18.0)	13.0 (9.00-20.0)	0.024
Time of ventilation (h)	19.0 (9.00-38.0)	17.0 (9.00-40.0)	0.842
Drainage output 24 h after surgery (mL)	360 (245-568)	350 (240-548)	0.666
Parameters of postoperative renal function			
Urine output 24 h after surgery (mL)	2,670 (2,199 to 3,252)	2,582 (2,145 to 3,172)	0.129
Percent change in creatinine on POD 1	8.00 (–4.84 to 24.1)	11.7 (–2.12 to 30.5)	0.005
Percent change in creatinine on POD 3	6.06 (–8.11 to 26.1)	9.09 (–6.08 to 34.7)	0.020
Percent change in creatinine on POD 4	4.23 (–10.06 to 24.1)	7.78 (–7.98 to 29.5)	0.010
Percent change in creatinine on POD 5	2.24 (–10.57 to 22.6)	6.37 (–8.54 to 29.1)	0.005
Percent change in creatinine on POD 6	2.29 (–10.55 to 22.2)	6.37 (–8.26 to 27.6)	0.006
Percent change in creatinine on POD 7	1.98 (–10.19 to 21.9)	6.62 (–8.42 to 27.6)	0.005
KDIGO			0.157
Stage 0	480 (80.3%)	444 (74.7%)	
Stage 1	73 (12.2%)	93 (15.7%)	
Stage 2	11 (1.84%)	14 (2.36%)	
Stage 3	34 (5.69%)	43 (7.24%)	
KDIGO (0 *v* 1 to 3)	118 (19.7%)	150 (25.3%)	0.027

Abbreviations: KDIGO, Kidney Disease: Improving Global Outcomes; POD, postoperative day.

more often in patients in the hypertensive group (25.3% *v* 19.7%, p = 0.027).

Robustness of the Results

To assess the robustness of the results, several sensitivity analyses were performed to confirm that the difference observed between the hypertensive and normotensive groups using the third quartile cut of AUC_{hyper} to define HTN was not by chance. To that end, when HTN was defined using different threshold criteria (ie, greater than third quartile cut of AUC_{hyper}, or 1,700 mmHg × min), similar results were found. An analysis using different threshold definitions for postoperative HTN (ie, AUC_{hyper} cut of >1,500; >2,000; and >2,500 mmHg × min) demonstrated consistent differences in outcome metrics between the normotensive and hypertensive groups (Fig 3).

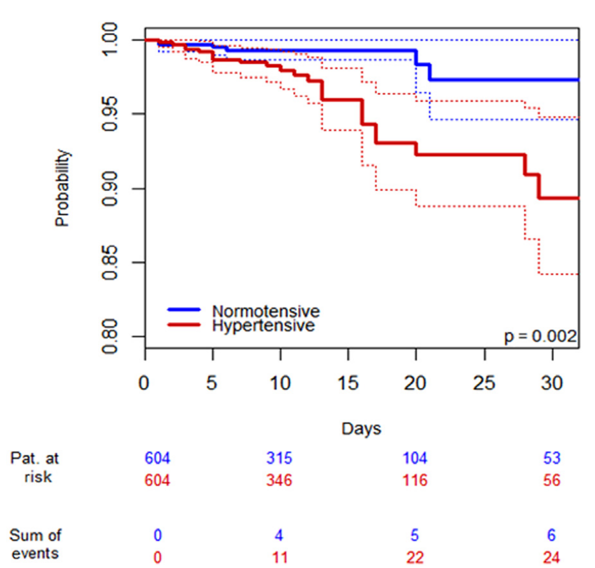

Fig 2. Survival analyses. Kaplan-Meier curves for in-hospital mortality of matched population. Pat., patients.

The relative odds for death in the hypertensive group at each AUC_{hyper} cut ranged from 3.38 to 4.7 compared with the relative odds for death in the normotensive group ($p \leq 0.001$; see Fig 3). Multivariate logistic regression and multivariate Cox proportional hazards regression on mortality adjusted for all preoperative/baseline parameters using backward stepwise selection were performed as additional sensitivity analyses. In all cuts and for both regression models, postoperative HTN had a significant impact on mortality.

Further sensitivity analysis was conducted at several cuts of the preoperative serum creatinine level to confirm the findings at different threshold criteria for renal insufficiency exclusion criteria. The greater relative risk of death in the hypertensive group remained when the exclusion threshold for the preoperative serum creatinine level was adjusted downward to 1.3 mg/dL (OR 2.44, 95% CI 1.18-5.43, p = 0.015) or upward to 1.7 mg/dL (OR 1.86, 95% CI 1.03-3.48, p = 0.041). The increases in the postoperative creatinine level over baseline on days 1 through 7 remained significantly different between groups under the same preoperative serum creatinine threshold variations (eTable 1).

As the capability to capture and record BP measurements postoperatively progressed to 24 hours and longer throughout the study, subgroup analyses, including patients who had at least 18 hours of data collection after ICU admission, were performed to account for possible selection bias. Matched pairs were found in the normotensive group for 542 patients (total 1,084—542 in each group). Again, it was observed that postoperative hypertensive patients had a significantly increased risk for mortality compared with normotensive patients (4.8% *v* 2.0%, p = 0.019). Consistent with these findings for the complete data set with BP measurements up to

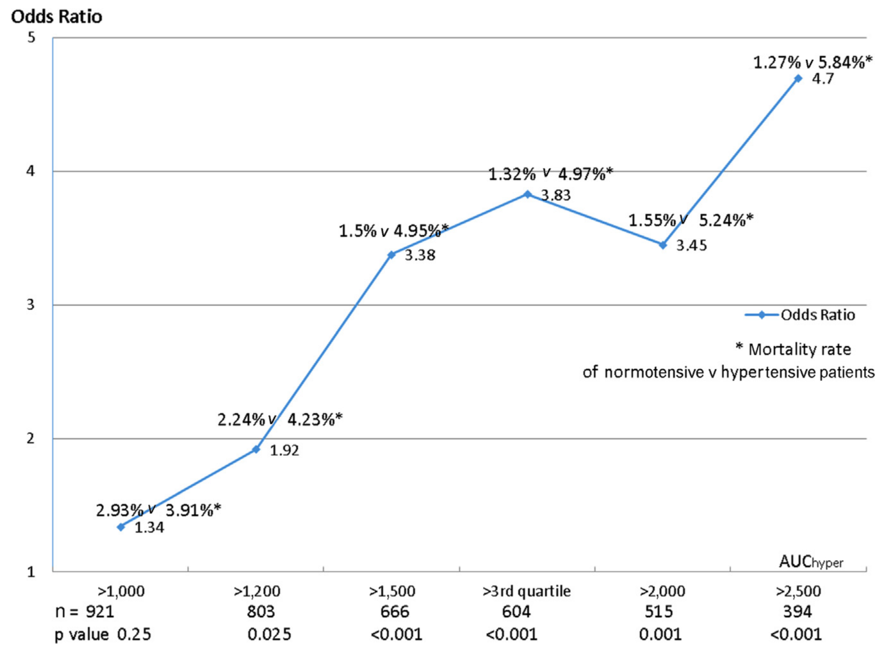

Fig 3. Sensitivity analyses. Odds ratio of mortality rate of normotensive versus hypertensive patients by AUC cut.

24 hours, both groups showed increased levels of serum creatinine postoperatively. Serum creatinine remained elevated in hypertensive patients and normalized significantly slower ($p < 0.05$ for group differences in percentage change from baseline after postoperative day 3) compared with normotensive patients.

DISCUSSION

Nearly 220 million patients around the world undergo surgery each year, with most requiring some form of deep sedation or general anesthesia, and are likely subjected to BP management during that time.[2] For this reason, BP monitoring during the perioperative period more than any other single parameter remains a core tenet of provider vigilance and BP management remains an important focus of perioperative clinical care. Despite this ubiquity, however, BP management considerations are not well-supported by a robust evidence base.[18,21]

The results of the study presented here suggested that the current, clinically acceptable guidelines for postoperative BP management after cardiac surgeries deserve re-examination because adverse outcomes were observed while adhering to commonly endorsed definitions and management strategies for postoperative HTN. Specifically, for patients who underwent cardiac surgery, a conservative definition of HTN within the first 24 hours after surgery was associated with significant AKI and increased 30-day mortality. The results of this study, which related early postoperative HTN to AKI, were consistent with a growing body of literature suggesting that poor BP control in the perioperative period can adversely affect both short- and long-term outcomes.[6–9]

This study's data set covered 7 years at a single major German academic medical center and included all patients presenting for cardiac surgery during this period. These analyses were important because they represented the first report specifically characterizing and implicating the adequacy of postoperative BP management on the development of adverse outcomes, including AKI and worsened mortality.

The following 3 aspects of this analysis merit further discussion: the context of the findings, the applicability of the findings to other surgical procedures and patient populations, and the nature of the data set used for this analysis.

Generally it is accepted that the concept of perioperative HTN—particularly in patients in the cardiac surgery setting—needs to be defined differently from that commonly referenced in an ambulatory environment due to unique clinical-physiologic considerations of anesthesia and surgery.[17,19] Most anesthesia providers generally consider some combination of patient demographics, preoperative morbidity, type of surgery, and nature of the anesthetic as components influencing BP management decisions and goals. Some effort has been made to quantitatively assess the nature of the relationship between perioperative hemodynamic parameters and surgical outcomes, but a rational methodology for choosing perioperative BP targets remains aspirational.[30,31] The results of this study suggested that a more empirical approach to these decisions is warranted.

Recent data in the ambulatory setting appropriately have drawn focus and challenge to traditional BP management standards.[32] Similar to ambulatory settings, the relationship between BP and outcomes in cardiovascular surgical settings remains poorly defined. Unlike the ambulatory setting, however, larger challenges lie in defining values for perioperative BP management and determining patient populations subject to those metrics. Similar to acute care settings,[33,34] the absolute degree of perisurgical HTN, hypotension, and variability all appear to contribute to short- and long-term postoperative outcomes.[4–6,11–13] Specific complications appear to relate to the applicability of BP targets across different patient types or surgical interventions.[11–13,35–37] The results of this study, showing a relationship between high postoperative BP with AKI and mortality, were confined to high-risk patients who underwent cardiac surgery. To what extent, if any, these results applied to patients with similar demographics who undergo noncardiac surgery or even to different patient cohorts remain speculative.

Understanding how BP values—whether the absolute value, variability, and/or deviation from baseline value—in the preoperative, intraoperative, and postoperative periods are related to one another is complex with mechanistic gaps remaining to be solved. For example, there is ample evidence that shows a clear relationship between both poor preoperative and intraoperative BP control and subsequent adverse outcomes, including renal impairment.[9,38,39] There clearly is underlying physiologic coupling between preoperative and postoperative BP. Indeed, 78.3% of patients in this study who were classified as experiencing postoperative HTN, presented with a history of preoperative HTN. In this context, this study's data suggested that although preoperative HTN may make postoperative HTN more likely or severe, the importance of management of BP in the postoperative period needs to be better appreciated and emphasized. This analysis offered little insight into the relationship between intraoperative BP and postoperative outcomes. The specifics of which intraoperative variables and by what mechanisms those variables might contribute to poor postoperative outcomes remain to be better elucidated.

After mortality, the primary adverse outcome reported in this analysis was AKI. Renal impairment after cardiac surgery is a serious complication. Whereas a large rise in serum creatinine level ($\geq$100%) over baseline has been shown to portend a doubling of in-hospital mortality, even small relative increases in creatinine can be used as a sensitive index of AKI after aortocoronary bypass graft surgery that is associated with higher mortality and substantial additive cost.[22,26,40–47] AKI is associated with preoperative HTN, increased pulse pressure, and intraoperative SBP variation.[5,7,9,12,13] Intraoperative SBP variation also predicts 30-day mortality.[5,6,12,13] The results of the study presented here added to this body of knowledge and showed for the first time that AKI was associated with early HTN after cardiac surgery. Even though AKI after cardiac surgery is associated with both short- and long-term mortality, this study's data set did not allow for the authors to argue for any potential causal relationship.[22,24,25]

The mechanisms responsible for how elevated BP may affect postoperative outcomes are an active area of research. Recent evidence suggests that for high-risk, albeit noncardiac surgical, patients, a generalized, low-level ischemic state may exist that is associated with subclinical myocardial injury. This injury pattern may account for as much as 34% of all 30-day postoperative surgical mortality in this population.[48] In addition, this ischemia and inflammation likely lead to injury in other organs. Similar to that which occurs in low-level myocardial injury, oxygen delivery–sensitive organs such as the brain and kidney also are prone to ischemic damage, implying that BP via its ultimate physiologic derivative of perfusion plays a major role in patterns and degree of postoperative organ injury.[49] Attempts to optimize perioperative hemodynamic management in high-risk surgical patients lessened the incidence and severity of postoperative AKI, further supporting the notion that perioperative BP control can be an important target for attenuating postsurgical AKI.[50–52]

This study had a number of important limitations that likely affected the general applicability and relevance of the findings across other demographics and surgical procedures. This analysis focused exclusively on postoperative HTN and renal outcomes. The authors did not provide any insight on the contribution of hypotension to the observed outcomes because the 1:1 matching criteria used for the hypertensive-to-normotensive comparison did not permit adequate-sized groups if applied to a hypotensive population. The authors simply would have had to establish less rigorous 1:1 matching criteria to compare a hypotensive with a normotensive population. The authors maintained that lack of a hypotensive comparative group likely did not influence the findings because the majority of patients who presented to the ICU were not hypotensive and because of the robustness of the matching parameters. Equally important, the authors were unable to provide any insight as to what, if any, intraoperative discriminators may be causally important to the observed postoperative outcomes. Given the known importance of BP control in the intraoperative period to postoperative outcomes, such as AKI, it is possible that the association of postoperative HTN with AKI after cardiac surgery may be related ultimately to events that occurred during the intraoperative period. The authors believed that their statistical analysis largely minimized this concern, but the potential confounding of intraoperative BP variables cannot be discounted. This analysis did not include other outcomes in addition to renal due to data availability. Even though this did not affect the interpretation of the results, it did leave open the possibility that AKI was not a primary outcome variable. It should be noted that the focus of this study was to compare a postoperative normotensive population after cardiac surgery with a postoperative population with a commonly used clinical threshold definition for HTN of similar characteristics (including well-accepted risk scores for ICU patients) to determine health outcome differences. The authors did not seek to test the robustness of one composite ICU risk score versus another.

Finally, as with all observational studies, the authors may not have been able to identify all relevant factors for the matching, regardless of the methodology used. Thus, analysis and inference may be subject to residual bias and unknown confounding due to unmeasurable factors.[53] Although all matching methods aim to balance the distribution of known covariates that may potentially affect outcomes (ie, confounders), no matching method can balance "unknown" confounders, and for this reason, no matching method is perfect. The authors used a 1:1 matching methodology over a propensity matching methodology for this data set specifically to minimize the effect of potential confounders and bias because unknown factors are not obviated by a propensity matching methodology. To this end, several matching methods were considered for this study to find optimal balance using the identified baseline covariates. The individual 1:1 matching method was chosen and performed with the R package "optmatch" version 0.9-1.[54] These standard differences between covariates are reported in eTable 2.

A disadvantage of individual matching, however, is to lose subjects when similar demographic and baseline characteristics for each use are not found. The more variables used, the chance of losing patients eligible for a matching comparative group is greater. That being said, the authors maintained that 68% (or 604 patients per group) was a very good representative sample in this population. Also, the sample size of 604 in each group provided >90% power to detect a mortality difference of 3.5% between groups.

APPENDIX A. SUPPLEMENTARY MATERIAL

Supplementary data associated with this article can be found in the online version at http://dx.doi.org/10.1053/j.jvca.2016.05.040.

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